Echo chambers: opinion against neighbours' opinion

An echo chamber means that agents mostly hear opinions like their own. To see whether that happens in our model, we compare each agent's opinion x with $\langle x \rangle^{NN}$, the mean opinion of the agents it heard (its nearest neighbours), as in Fig. 3a of Baumann et al. (2020).

The NetLogo model (`netlogo/polarisation.nlogox`, with `collect-snapshots?` on) records who heard whom over several time steps once the population is polarized, and writes one file per run with, among others, the columns `x` and `x_nn`. The Info tab of the model explains how. This notebook pools all runs of an m-mode and draws:

- in the main panel, the joint distribution of x and $\langle x \rangle^{NN}$ (bright = many agents). Echo chambers show up as two bright spots, bottom left and top right: agents with a negative opinion hear negative opinions, and agents with a positive opinion positive ones.
- above it, the distribution of the opinions, $P(x)$
- to the right, the distribution of the neighbours' mean opinions, $P_{NN}(x)$

Input and output:

- input: the .csv files in `data/snapshots/`, with one folder per m-mode (e.g. `data/snapshots/uniform/`) and one file per run
- output: one figure per m-mode in `results/snapshots/`, e.g. `results/snapshots/echo-chambers-uniform.png`

```
library(MASS) # for kde2d(), the 2-D density estimate

input_folder <- "../data/snapshots"
output_folder <- "../results/snapshots"
dir.create(output_folder, recursive = TRUE, showWarnings = FALSE)

modes <- c("uniform", "boost-extreme", "uprank-bridging-and-downrank-toxic", "uprank-bridging")
```

Settings

The same settings are used for every m-mode:

```
axis_limit <- 6 # both axes go from -6 to 6
bandwidth <- 0.25 # how much the densities are smoothed (standard deviation of the kernel)
grid_size <- 200 # the 2-D density is computed on a 200 x 200 grid
max_points <- 50000 # at most this many agents are used for the 2-D density
n_levels <- 15 # about this many colour levels
palette <- "Inferno"
```

Reading the runs

Each run file has one row per agent. We only need the opinion `x` and the neighbours' mean opinion `x_nn`, so we keep those two columns and stack the files of one m-mode on top of each other.

```
read_runs <- function(mode) {
  files <- list.files(file.path(input_folder, mode), pattern = "\\.csv$", full.names = TRUE)
  runs <- lapply(files, function(file) read.csv(file)[, c("x", "x_nn")])
  agents <- do.call(rbind, runs)
  cat(mode, ":", length(files), "runs,", nrow(agents), "agents\n")
  agents
}
```

Estimating the densities

The marginal densities $P(x)$ and $P_{NN}(x)$ use all agents. The 2-D density takes much more memory, so if there are more than `max_points` agents it uses a random sample of them. The seed makes that sample the same every time.

`kde2d()` and `density()` define the amount of smoothing differently: `density()` takes the standard deviation of the kernel, `kde2d()` four times that. We therefore pass `4 * bandwidth` to `kde2d()`, so that both are smoothed the same.

```
estimate_densities <- function(agents) {
  limits <- c(-axis_limit, axis_limit)

  set.seed(1)
  if (nrow(agents) > max_points) {
    sample_rows <- sample(nrow(agents), max_points)
  } else {
    sample_rows <- seq_len(nrow(agents))
  }

  joint <- kde2d(agents$x[sample_rows], agents$x_nn[sample_rows],
    n = grid_size, lims = c(limits, limits), h = rep(4 * bandwidth, 2)
  )

  list(
    joint = joint,
    x = density(agents$x, bw = bandwidth, from = -axis_limit, to = axis_limit),
    x_nn = density(agents$x_nn, bw = bandwidth, from = -axis_limit, to = axis_limit)
  )
}
```

Drawing the figure

The figure has three panels: the 2-D density in the main panel, $P(x)$ above it and $P_{NN}(x)$ to the right. `layout()` places them on a 2 x 2 grid: the numbers in its matrix say which panel goes where, with 0 for the empty top-right corner.

```
plot_echo_chambers <- function(densities, file) {
  limits <- c(-axis_limit, axis_limit)

  png(file, width = 1600, height = 1600, res = 200)
  layout(matrix(c(2, 0, 1, 3), nrow = 2, byrow = TRUE), widths = c(4, 1), heights = c(1, 4))

  # 1. main panel: filled contours of the 2-D density
```

```
  par(mar = c(4.5, 4.5, 0.5, 0.5))
  levels <- pretty(range(densities$joint$z), n_levels)
  plot.new()
  plot.window(xlim = limits, ylim = limits, xaxs = "i", yaxs = "i")
  .filled.contour(densities$joint$x, densities$joint$y, densities$joint$z,
    levels = levels, col = hcl.colors(length(levels) - 1, palette)
  )
  axis(1)
  axis(2)
  box()
  title(
    xlab = expression(italic(x)),
    ylab = expression(group(langle, italic(x), rangle)^NN)
  )

  # 2. top panel: P(x)
  par(mar = c(0, 4.5, 0.5, 0.5))
  plot(densities$x$x, densities$x$y,
    type = "l", lwd = 2,
    xlim = limits, xaxs = "i", xaxt = "n",
    xlab = "", ylab = expression(italic(P)(italic(x)))
  )

  # 3. right panel: P_NN(x), turned on its side
  par(mar = c(4.5, 0, 0.5, 0.5))
  plot(densities$x_nn$y, densities$x_nn$x,
    type = "l", lwd = 2,
    ylim = limits, yaxs = "i", yaxt = "n",
    xlab = expression(italic(P)[NN](italic(x))), ylab = ""
  )

  dev.off()
}
```

One figure per m-mode

```
files <- file.path(output_folder, paste0("echo-chambers-", modes, ".png"))

for (k in seq_along(modes)) {
  agents <- read_runs(modes[k])
  densities <- estimate_densities(agents)
  plot_echo_chambers(densities, files[k])
}
```

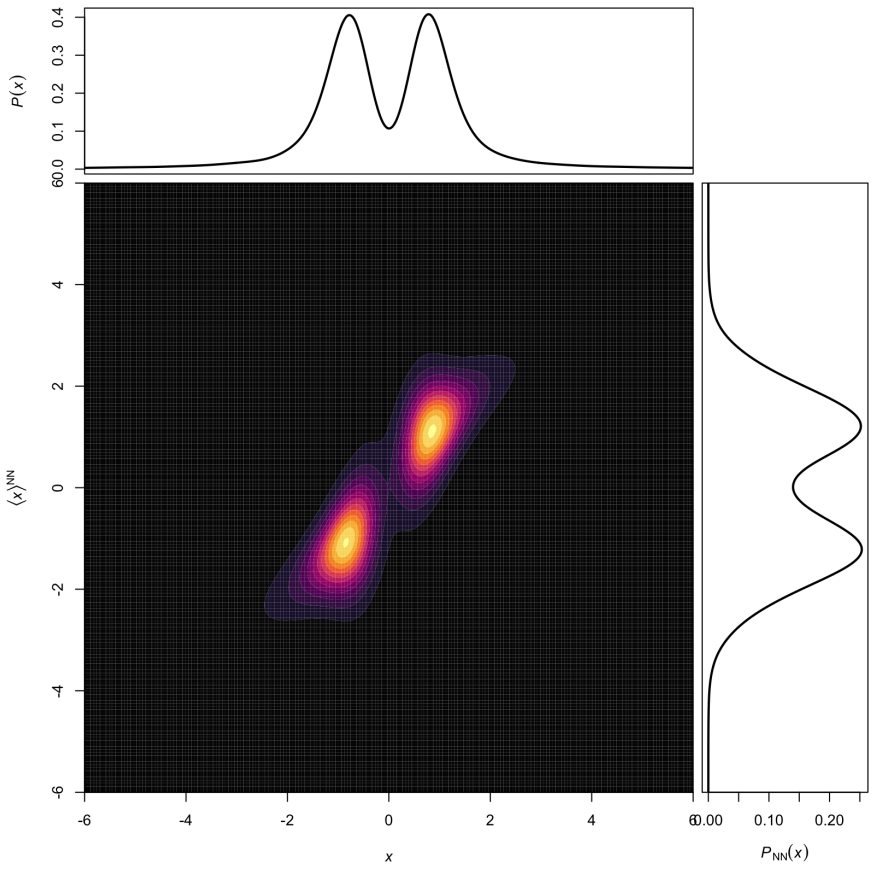
```
## uniform : 200 runs, 199999 agents

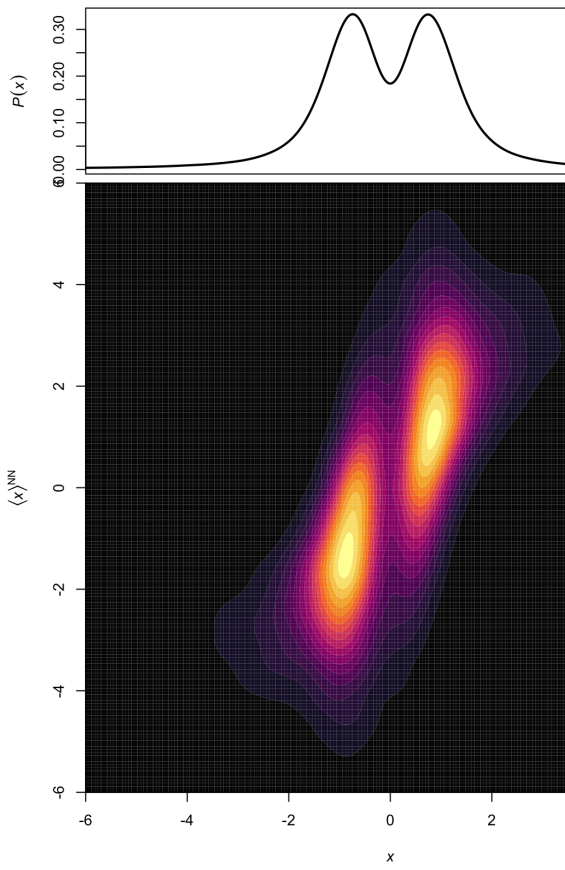
## boost-extreme : 200 runs, 200000 agents

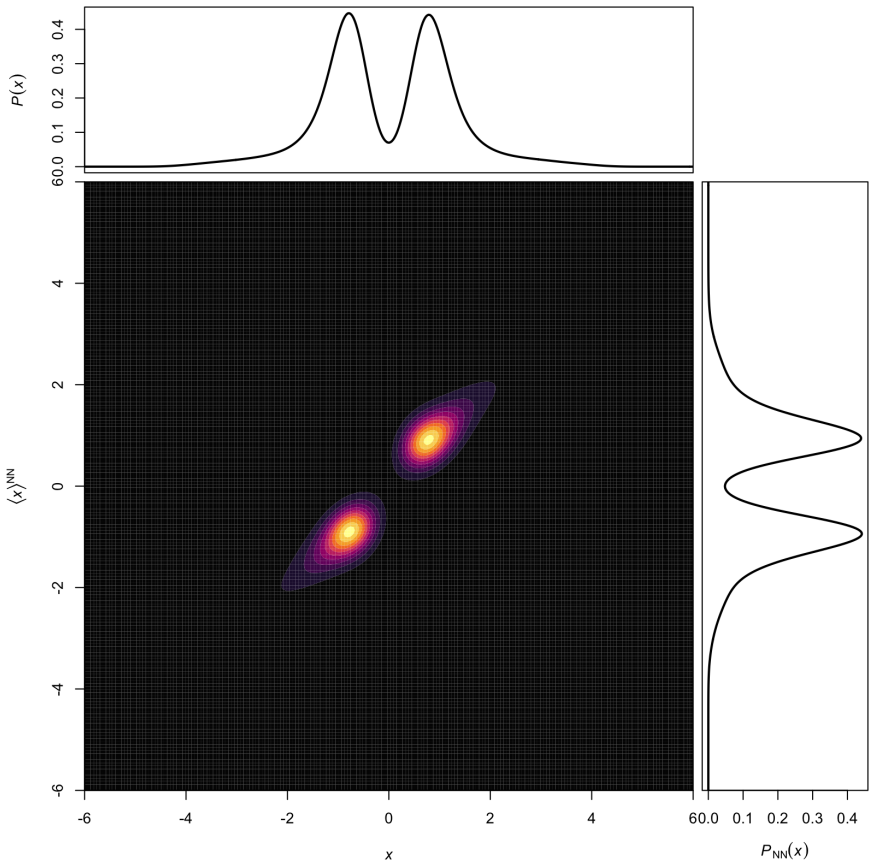
## uprank-bridging-and-downrank-toxic : 200 runs, 199941 agents

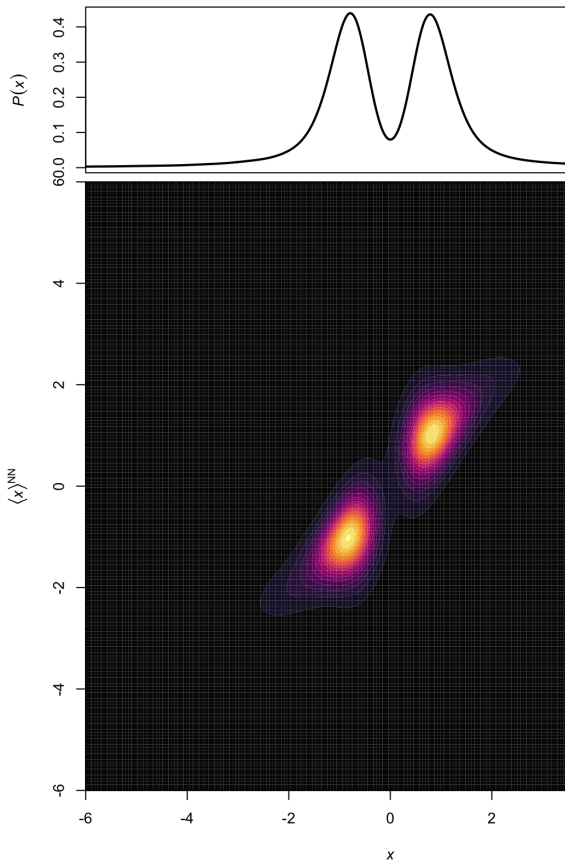
## uprank-bridging : 200 runs, 199993 agents
```

```
knitr::include_graphics(files)
```






P(x)
0.0 0.1 0.2 0.3 0.4
⟨x⟩NN
-6 -4 -2 0 2 4
x
PNN(x)


P(x)
0.0 0.1 0.2 0.3 0.4
⟨x⟩NN
-6 -4 -2 0 2
x